

T. de perturbaciones: método de proyectores

Expansión en serie de eigenvalores y eigenvectores derechos:

$$\lambda_n = \lambda_n^{(0)} + \epsilon \lambda_n^{(1)} + \epsilon^2 \lambda_n^{(2)} + \dots$$

$$\hat{\rho}_n = \hat{\rho}_n^{(0)} + \epsilon \hat{\rho}_n^{(1)} + \epsilon^2 \hat{\rho}_n^{(2)} + \dots$$

El superoperador de Lindblad perturbado es:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_1$$

Caso no degenerado

Sustituyendo esto en la ec. de eigenvalores obtenemos:

$$\epsilon^0: (\lambda_n^{(0)} - \mathcal{L}_0) \hat{\rho}_n^{(0)} = 0$$

$$\epsilon^1: (\lambda_n^{(0)} - \mathcal{L}_0) \hat{\rho}_n^{(1)} = (\mathcal{L}_1 - \lambda_n^{(1)}) \hat{\rho}_n^{(0)}$$

$$\epsilon^2: (\lambda_n^{(0)} - \mathcal{L}_0) \hat{\rho}_n^{(2)} = (\mathcal{L}_1 - \lambda_n^{(1)}) \hat{\rho}_n^{(1)} - \lambda_n^{(2)} \hat{\rho}_n^{(0)}$$

Definimos el proyector P_n como aquel que proyecta a una matriz de densidad cualquiera a su contribución en el n -ésimo eigenvector de orden cero.

$$P_n \rho = \text{Tr}(\hat{\rho}_n^{\dagger} \rho) \hat{\rho}_n^{(0)}$$

Por la relación de completitud el proyector Q_n que proyecta a todo el espacio excepto a la n -ésima contribución es:

$$Q_n \rho = \mathbb{1} \rho - P_n \rho$$

Ejemplos con proyectores para TLS:

$$P_1 |g\rangle\langle g| = \text{Tr}(\mathbb{1} |g\rangle\langle g|) |g\rangle\langle g| = |g\rangle\langle g|$$

$$P_2 |g\rangle\langle g| = \text{Tr}(|e\rangle\langle g| |g\rangle\langle g|) |g\rangle\langle g| = 0$$

$$P_3 |g\rangle\langle g| = \text{Tr}(|g\rangle\langle e| |g\rangle\langle g|) |e\rangle\langle g| = 0$$

$$P_4 |g\rangle\langle g| = \text{Tr}(|e\rangle\langle e| |g\rangle\langle g|) \sigma_z = 0$$

Aplicación de Proyectores

$$P_n \hat{\rho}_n^{(0)} = \hat{\rho}_n^{(0)}$$

$$P_n \hat{\rho}_n^{(p)} = 0 \quad \forall p > 0$$

$$Q_n \hat{\rho}_n^{(0)} = 0$$

$$Q_n \hat{\rho}_n^{(p)} = \hat{\rho}_n^{(p)} \quad \forall p > 0$$

$$P_n \rho = \text{Tr}(\hat{\rho}_n^{\dagger} \rho) \hat{\rho}_n^{(0)} \Rightarrow P_n |\rho\rangle\rangle = \langle\langle \hat{\rho}_n^{(0)} | \rho \rangle\rangle |\hat{\rho}_n^{(0)}\rangle\rangle \Rightarrow P_n = |\hat{\rho}_n^{(0)}\rangle\rangle \langle\langle \hat{\rho}_n^{(0)} |$$

$$P_n \hat{\rho}_n^{(p)} = \text{Tr}(\hat{\rho}_n^{\dagger} \hat{\rho}_n^{(p)}) \hat{\rho}_n^{(0)} = \begin{cases} \hat{\rho}_n^{(0)}, & \text{si } p=0 \\ 0, & \text{si } p \neq 0 \end{cases}, \quad Q_n \hat{\rho}_n^{(p)} = \hat{\rho}_n^{(p)} - P_n \hat{\rho}_n^{(p)} = \begin{cases} 0, & \text{si } p=0 \\ \hat{\rho}_n^{(p)}, & \text{si } p \neq 0 \end{cases}$$

Aplicamos P_n a la ec. de $O(\varepsilon)$:

$$P_n [(\lambda_n^{(0)} - \mathbb{I}_0) \hat{\beta}_n^{(1)}] = P_n [(\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(0)}] \Rightarrow P_n \mathbb{I}_1 \hat{\beta}_n^{(0)} = \lambda_n^{(1)} P_n \hat{\beta}_n^{(0)}$$

$$\Rightarrow \lambda_n^{(0)} \text{Tr}(\check{\beta}_n^{(0)\dagger} \hat{\beta}_n^{(1)}) \hat{\beta}_n^{(0)} - \text{Tr}(\check{\beta}_n^{(0)\dagger} \mathbb{I}_0 \hat{\beta}_n^{(1)}) \hat{\beta}_n^{(0)} \\ \hookrightarrow \text{Tr}[(\mathbb{I}_1 \check{\beta}_n^{(0)})^\dagger \hat{\beta}_n^{(1)}] = \text{Tr}[(\lambda_n^{(0)} \check{\beta}_n^{(0)})^\dagger \hat{\beta}_n^{(1)}] = \lambda_n^{(0)} \text{Tr}(\check{\beta}_n^{(0)\dagger} \hat{\beta}_n^{(1)})$$

$$= \text{Tr}(\check{\beta}_n^{(0)\dagger} \mathbb{I}_1 \hat{\beta}_n^{(0)}) \hat{\beta}_n^{(0)} - \lambda_n^{(1)} \text{Tr}(\check{\beta}_n^{(0)\dagger} \hat{\beta}_n^{(0)}) \hat{\beta}_n^{(0)}$$

$$\Rightarrow \lambda_n^{(1)} = \text{Tr}(\check{\beta}_n^{(0)\dagger} \mathbb{I}_1 \hat{\beta}_n^{(0)})$$

Aplicamos Q_n a la ec. de $O(\varepsilon)$:

$$Q_n [(\lambda_n^{(0)} - \mathbb{I}_0) \hat{\beta}_n^{(1)}] = Q_n [(\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(0)}]$$

$\mathbb{I}_0$ conmuta con P_n y Q_n :

$$\mathbb{I}_0 P_n \beta = \mathbb{I}_0 \text{Tr}(\check{\beta}_n^{(0)\dagger} \beta) \hat{\beta}_n^{(0)} = \text{Tr}(\check{\beta}_n^{(0)\dagger} \beta) \mathbb{I}_0 \hat{\beta}_n^{(0)} = \text{Tr}(\check{\beta}_n^{(0)\dagger} \beta) \lambda_n^{(0)} \hat{\beta}_n^{(0)} = \lambda_n^{(0)} \text{Tr}(\check{\beta}_n^{(0)\dagger} \beta) \hat{\beta}_n^{(0)}$$

$$P_n \mathbb{I}_0 \beta = \text{Tr}(\check{\beta}_n^{(0)\dagger} \mathbb{I}_0 \beta) \hat{\beta}_n^{(0)} = \text{Tr}[(\mathbb{I}_0^\dagger \check{\beta}_n^{(0)})^\dagger \beta] \hat{\beta}_n^{(0)} = \text{Tr}[(\lambda_n^{(0)} \check{\beta}_n^{(0)})^\dagger \beta] \hat{\beta}_n^{(0)} = \lambda_n^{(0)} \text{Tr}(\check{\beta}_n^{(0)\dagger} \beta) \hat{\beta}_n^{(0)}$$

$Q_n = \mathbb{I} - P_n$ y $\mathbb{I}$ conmuta con $\mathbb{I}_0$ así que Q_n también lo hace.

$$\Rightarrow Q_n \hat{\beta}_n^{(1)} = \frac{Q_n}{\lambda_n^{(0)} - \mathbb{I}_0} (\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(0)} \Rightarrow \hat{\beta}_n^{(1)} = \frac{Q_n}{\lambda_n^{(0)} - \mathbb{I}_0} (\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(0)} \\ \hookrightarrow \hat{\beta}_n^{(1)} = \text{Tr}(\check{\beta}_n^{(0)\dagger} \hat{\beta}_n^{(1)}) \hat{\beta}_n^{(0)}$$

$$\text{Si } \lambda_n^{(1)} = 0 \quad \forall n \Rightarrow \hat{\beta}_n^{(1)} = \frac{Q_n}{\lambda_n^{(0)} - \mathbb{I}_0} \mathbb{I}_1 \hat{\beta}_n^{(0)}$$

Ahora aplicamos P_n a la ec. de $O(\varepsilon^2)$:

$$P_n [(\lambda_n^{(0)} - \mathbb{I}_0) \hat{\beta}_n^{(2)}] = P_n [(\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(1)} - \lambda_n^{(2)} \hat{\beta}_n^{(0)}]$$

$$\Rightarrow (\lambda_n^{(0)} - \mathbb{I}_0) P_n \hat{\beta}_n^{(2)} = P_n (\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(1)} - \lambda_n^{(2)} P_n \hat{\beta}_n^{(0)}$$

$$\Rightarrow \lambda_n^{(2)} \hat{\beta}_n^{(0)} = P_n \mathbb{I}_1 \hat{\beta}_n^{(1)} = \text{Tr}(\check{\beta}_n^{(0)\dagger} \mathbb{I}_1 \hat{\beta}_n^{(1)}) \hat{\beta}_n^{(0)} \Rightarrow \lambda_n^{(2)} = \text{Tr}(\check{\beta}_n^{(0)\dagger} \mathbb{I}_1 \hat{\beta}_n^{(1)})$$

$$\Rightarrow \lambda_n^{(2)} = \text{Tr}(\check{\beta}_n^{(0)\dagger} \mathbb{I}_1 \frac{Q_n}{\lambda_n^{(0)} - \mathbb{I}_0} \mathbb{I}_1 \hat{\beta}_n^{(0)})$$

Finalmente aplicamos Q_n a la ec. de $O(\varepsilon^2)$:

$$Q_n [(\lambda_n^{(0)} - \mathbb{I}_0) \hat{\beta}_n^{(2)}] = Q_n [(\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(1)} - \lambda_n^{(2)} \hat{\beta}_n^{(0)}] \Rightarrow \hat{\beta}_n^{(2)} = \frac{Q_n}{\lambda_n^{(0)} - \mathbb{I}_0} \mathbb{I}_1 \hat{\beta}_n^{(1)}$$

$$\Rightarrow \hat{\beta}_n^{(2)} = \frac{Q_n}{\lambda_n^{(0)} - \mathbb{I}_0} \mathbb{I}_1 \frac{Q_n}{\lambda_n^{(0)} - \mathbb{I}_0} \mathbb{I}_1 \hat{\beta}_n^{(0)}$$

Caso degenerado

Aquí $\lambda_n^{(0)} = \lambda_D^{(0)} \quad \forall n \in D \rightarrow$ eigenvalores degenerados

$$\mathcal{E}^0: (\lambda_D^{(0)} - \mathcal{L}_0) \hat{\mathcal{F}}_n^{(0)} = 0$$

$$\mathcal{E}^1: (\lambda_D^{(0)} - \mathcal{L}_0) \hat{\mathcal{F}}_n^{(1)} = (\mathcal{L}_1 - \lambda_n^{(1)}) \hat{\mathcal{F}}_n^{(0)}$$

$$\mathcal{E}^2: (\lambda_D^{(0)} - \mathcal{L}_0) \hat{\mathcal{F}}_n^{(2)} = (\mathcal{L}_1 - \lambda_n^{(1)}) \hat{\mathcal{F}}_n^{(1)} - \lambda_n^{(2)} \hat{\mathcal{F}}_n^{(0)}$$

$$\mathcal{E}^3: (\lambda_D^{(0)} - \mathcal{L}_0) \hat{\mathcal{F}}_n^{(3)} = (\mathcal{L}_1 - \lambda_n^{(1)}) \hat{\mathcal{F}}_n^{(2)} - \lambda_n^{(2)} \hat{\mathcal{F}}_n^{(1)} - \lambda_n^{(3)} \hat{\mathcal{F}}_n^{(0)}$$

$$\mathcal{E}^4: (\lambda_D^{(0)} - \mathcal{L}_0) \hat{\mathcal{F}}_n^{(4)} = (\mathcal{L}_1 - \lambda_n^{(1)}) \hat{\mathcal{F}}_n^{(3)} - \lambda_n^{(2)} \hat{\mathcal{F}}_n^{(2)} - \lambda_n^{(3)} \hat{\mathcal{F}}_n^{(1)} - \lambda_n^{(4)} \hat{\mathcal{F}}_n^{(0)}$$

Definimos los proyectores en el subespacio degenerado:

$P_D = \sum_{m \in D} P_m$, una suma sobre los proyectores a los eigenvectores degenerados porque todos ellos crean la singularidad $\frac{1}{\lambda_D - \mathcal{L}_0}$, así Q_D proyecta a un subespacio sin λ_D .

$$Q_D = \mathbb{1} - P_D$$

Veamos como actúan estos proyectores sobre los eigenvectores de $O(\mathcal{E})$:

$$P_D \hat{\mathcal{F}}_n^{(0)} = \sum_{m \in D} P_m \hat{\mathcal{F}}_n^{(0)} = \sum_{m \in D} \text{Tr}(\hat{\mathcal{F}}_m^{(0)\dagger} \hat{\mathcal{F}}_n^{(0)}) \hat{\mathcal{F}}_m^{(0)} = \sum_{m \in D} \delta_{mn} \hat{\mathcal{F}}_n^{(0)} = \hat{\mathcal{F}}_n^{(0)}$$

$$P_D \hat{\mathcal{F}}_n^{(1)} = \sum_{m \in D} P_m \hat{\mathcal{F}}_n^{(1)} = \sum_{m \in D} \text{Tr}(\hat{\mathcal{F}}_m^{(0)\dagger} \hat{\mathcal{F}}_n^{(1)}) \hat{\mathcal{F}}_m^{(0)}$$

$$Q_D \hat{\mathcal{F}}_n^{(0)} = \hat{\mathcal{F}}_n^{(0)} - P_D \hat{\mathcal{F}}_n^{(0)} = 0$$

$$Q_D \hat{\mathcal{F}}_n^{(1)} = \hat{\mathcal{F}}_n^{(1)} - \sum_{m \in D} \text{Tr}(\hat{\mathcal{F}}_m^{(0)\dagger} \hat{\mathcal{F}}_n^{(1)}) \hat{\mathcal{F}}_m^{(0)}$$

La aplicación de los proyectores a las correcciones no se simplifican, debemos encontrarlas ($P > 0$)

• Aplicamos P_D a la ec. de $O(\mathcal{E})$:

$$P_D [(\lambda_D^{(0)} - \mathcal{L}_0) \hat{\mathcal{F}}_n^{(1)}] = P_D [(\mathcal{L}_1 - \lambda_n^{(1)}) \hat{\mathcal{F}}_n^{(0)}] \Rightarrow \lambda_n^{(1)} \underbrace{P_D \hat{\mathcal{F}}_n^{(0)}}_{\hat{\mathcal{F}}_n^{(0)}} = P_D \mathcal{L}_1 \hat{\mathcal{F}}_n^{(0)}$$

$$\Rightarrow \lambda_n^{(1)} \hat{\mathcal{F}}_n^{(0)} = P_D \mathcal{L}_1 P_D \hat{\mathcal{F}}_n^{(0)} = \sum_{m \in D} \text{Tr}(\hat{\mathcal{F}}_m^{(0)\dagger} \mathcal{L}_1 \hat{\mathcal{F}}_n^{(0)}) \hat{\mathcal{F}}_m^{(0)}$$

Para 2 TLS $P_D \mathcal{L}_1 P_D = 0 \Rightarrow \lambda_D^{(1)} = 0$

• Aplicamos Q_D a la ec. de $O(\mathcal{E})$:

$$Q_D [(\lambda_D^{(0)} - \mathcal{L}_0) \hat{\mathcal{F}}_n^{(1)}] = Q_D [(\mathcal{L}_1 - \lambda_n^{(1)}) \hat{\mathcal{F}}_n^{(0)}] \Rightarrow (\lambda_D^{(0)} - \mathcal{L}_0) Q_D \hat{\mathcal{F}}_n^{(1)} = Q_D \mathcal{L}_1 \hat{\mathcal{F}}_n^{(0)} - \lambda_n^{(1)} Q_D \hat{\mathcal{F}}_n^{(0)}$$

$$\Rightarrow Q_D \hat{\mathcal{F}}_n^{(1)} = \underbrace{\frac{Q_D}{\lambda_D^{(0)} - \mathcal{L}_0}}_{S_D} \mathcal{L}_1 \hat{\mathcal{F}}_n^{(0)} \Rightarrow Q_D \hat{\mathcal{F}}_n^{(1)} = S_D \mathcal{L}_1 \hat{\mathcal{F}}_n^{(0)}$$

• Aplicamos P_0 a la ec. de $O(\epsilon^2)$: $\hat{\beta}_n^{(0)}$ $\rightarrow P_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(1)} + P_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(1)}$

$$P_0 (\lambda_0^{(0)} - \mathbb{I}_0) \hat{\beta}_n^{(1)} = P_0 (\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(1)} - \lambda_n^{(2)} P_0 \hat{\beta}_n^{(0)} \Rightarrow P_0 \mathbb{I}_1 \hat{\beta}_n^{(1)} = \lambda_n^{(2)} \hat{\beta}_n^{(0)}$$

Sustituyendo $\hat{\beta}_n^{(1)}$ obtenemos:

$$P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(0)} = \lambda_n^{(2)} \hat{\beta}_n^{(0)} \Rightarrow \boxed{M^{(2)} \hat{\beta}_n^{(0)} = \lambda_n^{(2)} \hat{\beta}_n^{(0)}}$$

cc. de eigenvalores para obtener las correcciones a 2º orden de eigenvalores y la buena base de eigenvectores a orden cero

• Aplicamos Q_0 a la ec. de $O(\epsilon^2)$:

$$Q_0 (\lambda_0^{(0)} - \mathbb{I}_0) \hat{\beta}_n^{(1)} = Q_0 (\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(1)} - \lambda_n^{(2)} Q_0 \hat{\beta}_n^{(0)}$$

$$\Rightarrow (\lambda_0^{(0)} - \mathbb{I}_0) Q_0 \hat{\beta}_n^{(1)} = Q_0 \mathbb{I}_1 \hat{\beta}_n^{(1)} \Rightarrow Q_0 \hat{\beta}_n^{(2)} = S_0 \mathbb{I}_1 \hat{\beta}_n^{(0)}$$

$$\Rightarrow Q_0 \hat{\beta}_n^{(2)} = S_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(1)} + S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(1)} \leftarrow \text{separamos } \hat{\beta}_n^{(1)} \text{ en su contribución en el sub-espacio no-degenerado y el sub-espacio degenerado}$$

Debemos hallar $Q_0 \hat{\beta}_n^{(1)}$ y $P_0 \hat{\beta}_n^{(1)}$.

Multiplicamos la ec. anterior por $P_0 \mathbb{I}_1$:

$$\Rightarrow P_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(2)} = P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(1)} + P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(1)}$$

• Aplicamos P_0 a la ec. de $O(\epsilon^3)$:

$$P_0 (\lambda_0^{(0)} - \mathbb{I}_0) \hat{\beta}_n^{(2)} = P_0 (\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(2)} - P_0 \lambda_n^{(2)} \hat{\beta}_n^{(1)} - P_0 \lambda_n^{(3)} \hat{\beta}_n^{(0)}$$

$$\Rightarrow P_0 \mathbb{I}_1 \hat{\beta}_n^{(2)} = \lambda_n^{(2)} P_0 \hat{\beta}_n^{(1)} + \lambda_n^{(3)} \hat{\beta}_n^{(0)} \Rightarrow \underbrace{P_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(2)} + P_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(2)}}_0 = \lambda_n^{(2)} P_0 \hat{\beta}_n^{(1)} + \lambda_n^{(3)} \hat{\beta}_n^{(0)}$$

$$\Rightarrow P_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(2)} = \lambda_n^{(2)} P_0 \hat{\beta}_n^{(1)} + \lambda_n^{(3)} \hat{\beta}_n^{(0)}, \text{ Igualamos estas ecuaciones:}$$

$$\Rightarrow P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(1)} + P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(1)} = \lambda_n^{(2)} P_0 \hat{\beta}_n^{(1)} + \lambda_n^{(3)} \hat{\beta}_n^{(0)}$$

$$\Rightarrow P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(1)} - \lambda_n^{(3)} \hat{\beta}_n^{(0)} = (\lambda_n^{(2)} - P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0) P_0 \hat{\beta}_n^{(1)} \rightarrow P_0^2 = P_0 \text{ (idempotentes)}$$

$$\Rightarrow P_0 \hat{\beta}_n^{(1)} = \frac{P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(1)} \rightarrow S_0 \mathbb{I}_1 \hat{\beta}_n^{(0)}}{\lambda_n^{(2)} - \underbrace{P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0}_{M^{(2)}}} - \frac{\lambda_n^{(3)} \hat{\beta}_n^{(0)}}{\lambda_n^{(2)} - P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0}$$

$$= \frac{1}{\lambda_n^{(2)} - M^{(2)}} P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(0)} - \frac{1}{\lambda_n^{(2)} - M^{(2)}} \lambda_n^{(3)} \hat{\beta}_n^{(0)}$$

Podemos tener singularidad, para evitarlo multiplicamos por Q_n (no Q_0 ya que $\lambda_n^{(2)}$ no es degenerado).

$$Q_n P_0 \hat{\beta}_n^{(1)} = \frac{Q_n}{\lambda_n^{(2)} - M^{(2)}} P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(0)} - \frac{\lambda_n^{(3)}}{\lambda_n^{(2)} - M^{(2)}} Q_n \hat{\beta}_n^{(0)}$$

$\rightarrow P_0 Q_n \hat{\beta}_n^{(1)} = P_0 \hat{\beta}_n^{(0)}$

$$Q_n P_0 = (1 - P_n) \sum_{m \in O} P_m = \sum_{m \in O} P_m - \sum_{m \in O} P_n P_m \quad \rangle \quad P_n P_m = P_m P_n \Rightarrow [Q_n, P_0] = 0$$

$$P_0 Q_n = \sum_{m \in O} P_m (1 - P_n) = \sum_{m \in O} P_m - \sum_{m \in O} P_m P_n$$

$$\Rightarrow P_0 \hat{\beta}_n^{(1)} = \frac{Q_n}{\lambda_n^{(2)} - M^{(2)}} P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(0)} = 0 \text{ en 2 TLS}$$

$$\Rightarrow Q_D \hat{\beta}_n^{(2)} = S_0 \mathbb{I}_1 S_0 \mathbb{I}_1 \hat{\beta}_n^{(0)}$$

• Aplicamos P_0 a la ec. de $O(\varepsilon^4)$:

$$P_0 (\lambda_0^{(0)} - \mathbb{I}_0) \hat{\beta}_n^{(4)} = P_0 (\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(3)} - P_0 \lambda_n^{(2)} \hat{\beta}_n^{(2)} - P_0 \lambda_n^{(3)} \hat{\beta}_n^{(1)} - P_0 \lambda_n^{(4)} \hat{\beta}_n^{(0)}$$

$$\Rightarrow P_0 \mathbb{I}_1 Q_D \hat{\beta}_n^{(3)} = \lambda_n^{(2)} P_0 \hat{\beta}_n^{(2)} + \lambda_n^{(3)} P_0 \hat{\beta}_n^{(1)} + \lambda_n^{(4)} \hat{\beta}_n^{(0)}$$

• Aplicamos Q_0 a la ec. de $O(\varepsilon^3)$:

$$Q_0 (\lambda_0^{(0)} - \mathbb{I}_0) \hat{\beta}_n^{(3)} = Q_0 (\mathbb{I}_1 - \lambda_n^{(1)}) \hat{\beta}_n^{(2)} - Q_0 \lambda_n^{(2)} \hat{\beta}_n^{(1)}$$

$$\begin{aligned} \Rightarrow Q_0 \hat{\beta}_n^{(3)} &= S_0 \mathbb{I}_1 \hat{\beta}_n^{(2)} - \lambda_n^{(2)} S_0 \hat{\beta}_n^{(1)} \\ &= S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(2)} + S_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(2)} - \lambda_n^{(2)} S_0 \hat{\beta}_n^{(1)} \end{aligned}$$

Sustituyendo $Q_0 \hat{\beta}_n^{(3)}$ en el resultado anterior:

$$P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(2)} + P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(2)} - \lambda_n^{(2)} P_0 \mathbb{I}_1 S_0 \hat{\beta}_n^{(1)} = \lambda_n^{(2)} P_0 \hat{\beta}_n^{(2)} + \lambda_n^{(3)} P_0 \hat{\beta}_n^{(1)} + \lambda_n^{(4)} \hat{\beta}_n^{(0)}$$

$$\Rightarrow P_0 \hat{\beta}_n^{(2)} = \frac{P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(2)} - \lambda_n^{(2)} P_0 \mathbb{I}_1 S_0 \hat{\beta}_n^{(1)} - \lambda_n^{(3)} P_0 \hat{\beta}_n^{(1)} - \lambda_n^{(4)} \hat{\beta}_n^{(0)}}{\lambda_n^{(2)} - M^{(2)}}$$

De nuevo multiplicamos por Q_n para evitar la singularidad:

= 0 en 2 TLS

$$\begin{aligned} \underbrace{Q_n P_0 \hat{\beta}_n^{(2)}}_{P_0 \hat{\beta}_n^{(2)}} &= \frac{Q_n}{\lambda_n^{(2)} - M^{(2)}} P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 Q_0 \hat{\beta}_n^{(2)} - \frac{Q_n}{\lambda_n^{(2)} - M^{(2)}} \lambda_n^{(2)} P_0 \mathbb{I}_1 S_0 \hat{\beta}_n^{(1)} - \frac{\lambda_n^{(3)} Q_n}{\lambda_n^{(2)} - M^{(2)}} \underbrace{P_0 \hat{\beta}_n^{(1)}}_{=0 \text{ en 2 TLS}} \\ &\quad - \frac{\lambda_n^{(4)} Q_n}{\lambda_n^{(2)} - M^{(2)}} \hat{\beta}_n^{(0)} \quad \lambda_n^{(2)} P_0 \mathbb{I}_1 S_0 \hat{\beta}_n^{(1)} = \lambda_n^{(2)} P_0 \mathbb{I}_1 S_0 S_0 \mathbb{I}_1 \hat{\beta}_n^{(0)} = \frac{\lambda_n^{(2)}}{\lambda_0 - \mathbb{I}_0} P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 P_0 \hat{\beta}_n^{(0)} \\ &\quad = \frac{\lambda_n^{(2)}}{\lambda_0 - \mathbb{I}_0} M^{(2)} \hat{\beta}_n^{(0)} = \frac{\lambda_n^{(2)^2}}{\lambda_0 - \mathbb{I}_0} \hat{\beta}_n^{(0)} \end{aligned}$$

$$\Rightarrow \frac{Q_n}{\lambda_n^{(2)} - M^{(2)}} \lambda_n^{(2)} P_0 \mathbb{I}_1 S_0 \hat{\beta}_n^{(1)} = \frac{\lambda_n^{(2)^2}}{(\lambda_n^{(2)} - M^{(2)})(\lambda_0 - \mathbb{I}_0)} Q_n \hat{\beta}_n^{(0)} = 0$$

$$\Rightarrow P_0 \hat{\beta}_n^{(2)} = \frac{Q_n}{\lambda_n^{(2)} - M^{(2)}} P_0 \mathbb{I}_1 S_0 \mathbb{I}_1 S_0 \mathbb{I}_1 S_0 \mathbb{I}_1 \hat{\beta}_n^{(0)}$$